

Ayush Lahiri

Software/M.L. Engineer

+91-8340708543 | ayush.lahiri.dev@gmail.com | LinkedIn | GitHub | Leetcode | Portfolio | Kolkata, West Bengal

SUMMARY

CSE undergrad building ML systems from first principles — implemented a **55M-parameter Transformer** and a custom autograd engine from scratch, alongside fine-tuned NLP models. Solved 230+ problem and **Leetcode Knight**.

EDUCATION

IIIT Kalyani <i>B.Tech CSE, CGPA: 8.24 (absolute)</i>	Sept 2025 – Jun 2029
Mothers' Intl. Academy, 12th <i>CBSE, Percentage: 78%</i>	Apr 2023 – Apr 2025
Carmel School, 10th <i>CISCE, Percentage: 93.4%</i>	Apr 2014 – Apr 2023

TECHNICAL SKILLS

Languages: Python, C, Java [Familiar With: C++]
Developer Tools: VS Code, Android Studio, Git/GitHub, Jupyter Notebook/Lab
Frameworks: Numpy, Pandas, PyTorch, Docker, Hugging Face
DevOps: Docker, Git/GitHub, CI/CD Basics, Linux
Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Architecture & Operating Systems

PROJECTS

Transformer LM <i>PyTorch, Numpy, Python/C++</i> – Implemented a ~55(54.619984) Million parameter Transformer LM from scratch using Numpy and PyTorch, trained on the 1M-TinyStories Dataset . – Additionally trained a BPE 10K-Context window Tokenizer for the Transformer from scratch. – Used core-level multiprocessing to increase training throughput by ~4% .	Jun '26 – Aug '26
Medical Emergency System Github <i>Node.js, PyTorch, Numpy, MongoDB, Hugging Face</i> – Engineered a specialised Medical Emergency Response System based on patient history, specialty, and availability of ambulance and hospital. – Fine-Tuned gLiner-Largev2.1 on triage data for description-to-emergency classification with 20% more accuracy & 6x faster than manual allocation. – Implemented cluster-based ruling to allocate hospital and ambulance using a neural network and LSTM (trained on synthetic data).	Jun '26 – Jun '26
Autograd Engine Github <i>Python</i> – Designed and implemented an Autograd engine from scratch with secondary data-structure and complex-ops support. – Extended the engine to implement a MultiLayered Linear Regression model. – Designed and implemented Softmax and Sigmoid functions and a sample Neural Network.	Apr '26 – May '26

EXPERIENCE

Lead of FOSS Club IIIT Kalyani <i>IIIT, Kalyani</i> – Conducted Core Team Onboarding. – Spearheaded a technical workshop for 70+ participants .	Sept 2026 – Present
App-Dev Lead of GDG IIITK <i>IIIT, Kalyani</i> – Designed and implemented user interfaces for Community Event that engaged 80+ users and produced immersive demo experiences across sessions. – Spearheaded a technical workshop on the app development lifecycle for 80+ participants . – Conducted GDG IIITK's flagship event WOC along with fellow leads.	Oct 2025 – Present
Open Source Contributions <i>Apetre 3.0</i> – Implemented Visual and Audio Alert for the Drowsiness Detector with 10ms latency .	Oct 2025 – Nov 2025

ACHIEVEMENTS & CERTIFICATIONS

- **1900+(knight)** on Leetcode & solved **230+** coding problems.
- Finalist in multiple hackathons (**Innovate-X, BotBalls-Top4, Code Canvas**).
- Achieved **Top 4%** in **EROH RING OF HONOUR** by **Naukri.com**.
- Completed **CS229** (Machine Learning), **6.086** (Hands-on Deep Learning).